

The Physics of Fluids 2: Buoyancy and Archimedes' Principle

Introduction: The Invisible Force That Makes Objects Float

Why does a massive steel ship float while a small iron nail sinks? Why do people feel lighter when submerged in water? Why do icebergs float with only a fraction of their mass visible above the surface? These phenomena are all explained by a single principle discovered over two thousand years ago — **Archimedes' Principle**.

When an object is placed in a fluid (a liquid or gas), the fluid pushes back. This upward push is called the **buoyant force**. Archimedes' Principle states that the buoyant force acting on an object is equal to the weight of the fluid displaced by the object. In simpler terms, when an object is submerged, it pushes aside (displaces) a certain amount of fluid. The weight of that displaced fluid determines how strongly the fluid pushes back upward.

This principle, combined with the concept of **density** (mass per unit volume), explains why some objects float while others sink. An object floats if its density is less than the density of the fluid; it sinks if its density is greater. Shape also matters — a flat sheet of metal may float while the same mass of metal crumpled into a ball may sink, because the shape affects how much fluid is displaced.

This reading material examines Archimedes' Principle, the relationship between buoyancy and density, the factors that determine whether an object floats or sinks, and the real-world applications of these concepts in shipbuilding, submarines, hot-air balloons, and hydrometers.

Part 1: Buoyancy — The Upward Push of Fluids

Definition of Buoyancy

Buoyancy is the upward force exerted by a fluid that opposes the weight of an object immersed in the fluid. This force is the reason objects feel lighter in water and why some objects float.

The Feeling of "Lighter" in Water

When a person lifts a heavy object underwater, it feels significantly lighter than when lifted in air. This is not because the object's actual weight has changed, but because water exerts an upward buoyant force that counteracts part of the object's weight. The same principle applies to lifting a friend in a swimming pool — the water pushes up, making the person easier to lift.

Demonstration: Weight in Air vs. Weight in Water

If a spring scale is used to measure the weight of a heavy stone in air, the scale shows the stone's full weight. When the same stone is submerged in water, the scale shows a smaller reading. The difference between the two readings is the buoyant force exerted by the water.

Measurement	Value
Weight in air	10 N
Weight in water	6 N
Buoyant force	4 N (10 N - 6 N)

Part 2: Archimedes' Principle

The Story of Archimedes' Discovery

According to historical accounts, the Greek mathematician Archimedes was tasked with determining whether a crown was made of pure gold or contained silver, without damaging the crown. While stepping into a bath, he noticed that the water level rose as his body entered the tub. He realized that the volume of water displaced was equal to the volume of his body submerged. Excited by his discovery, he reportedly ran through the streets shouting "Eureka!" — Greek for "I have found it!"

Statement of Archimedes' Principle

Archimedes' Principle states that the buoyant force acting on an object immersed in a fluid is equal to the weight of the fluid displaced by the object.

Mathematically:

$$F_b = \rho_{\text{fluid}} \times V_{\text{displaced}} \times g$$

Where:

- F_b = buoyant force (in Newtons, N)
- ρ_{fluid} = density of the fluid (in kg/m^3)
- $V_{\text{displaced}}$ = volume of fluid displaced (in m^3)
- g = acceleration due to gravity (9.8 m/s^2)

The Relationship Between Weight Loss and Displaced Fluid

When an object is submerged, it loses apparent weight equal to the weight of the fluid it displaces. This "weight loss" is exactly the buoyant force.

Observation

Explanation

Object weighs 10 N in air

Actual weight

Observation	Explanation
Object weighs 6 N in water	Apparent weight
Weight loss = 4 N	This equals the weight of water displaced
Volume of water displaced	Equal to the volume of the object submerged

Part 3: Density — The Deciding Factor for Floating and Sinking

Definition of Density

Density is the mass of an object per unit volume. It is a measure of how tightly packed the matter in an object is.

Formula:

$$\rho = m / V$$

Where:

- ρ (rho) = density (in kg/m³ or g/cm³)
- m = mass (in kg or g)
- V = volume (in m³ or cm³)

Comparing Densities: The Key to Floating or Sinking

Condition	Result	Example
Object density < fluid density	Object floats	Wood (density ~0.6 g/cm ³) in water (density 1.0 g/cm ³)

Condition	Result	Example
Object density = fluid density	Object remains suspended at any depth	Some submarines, neutral buoyancy
Object density > fluid density	Object sinks	Iron (density $\sim 7.9 \text{ g/cm}^3$) in water (density 1.0 g/cm^3)

Density of Common Substances

Substance	Density (g/cm^3)	Floats or Sinks in Water?
Air	0.0012	Floats (but rises through water as bubbles)
Cork	0.24	Floats
Ice	0.92	Floats
Water	1.00	(reference)
Human body (average)	~ 0.98	Usually floats with effort
Aluminum	2.70	Sinks
Iron	7.87	Sinks
Gold	19.30	Sinks

Why Ice Floats on Water

Ice has a density of approximately 0.92 g/cm^3 , while liquid water has a density of 1.00 g/cm^3 . Because ice is less dense than water, it floats. Approximately 92% of an iceberg's volume is submerged, with only about 8% visible above the surface. This is why the phrase "tip of the iceberg" refers to a small visible part of a much larger hidden mass.

Part 4: The Role of Shape in Floating

Why a Steel Ship Floats While a Steel Nail Sinks

A solid steel nail has a density much greater than water (7.9 g/cm^3 vs. 1.0 g/cm^3), so it sinks. However, a steel ship is not solid — it is hollow. The ship's hull encloses a large volume of air, which significantly reduces the overall density of the ship. The average density of the ship (mass of steel plus mass of enclosed air divided by total volume) becomes less than the density of water, allowing the ship to float.

Object	Mass	Volume	Density	Floats?
Steel nail	High (for its size)	Small	High ($> 1.0\text{ g/cm}^3$)	No (sinks)
Steel ship (hollow)	High (but large volume)	Large	Low ($< 1.0\text{ g/cm}^3$)	Yes (floats)

How Shape Affects Displacement and Buoyancy

The shape of an object determines how much water it displaces. A flat, wide shape displaces more water than a compact, narrow shape of the same mass. More displacement means greater buoyant force.

Shape	Displacement	Buoyant Force	Result
Flat foil sheet	Large	Large	Floats
Crumpled foil ball	Small	Small	Sinks

This explains why a flat sheet of aluminum foil floats, but the same foil crumpled into a tight ball sinks — the mass is the same, but the volume and shape have changed, altering the displacement and therefore the buoyant force.

Part 5: Applications of Archimedes' Principle

Ships and Submarines

Vessel	How It Uses Archimedes' Principle
Ship	The hollow hull displaces a large volume of water, creating a buoyant force equal to the ship's weight. The ship floats as long as its average density is less than water.
Submarine	Ballast tanks are filled with water to increase density and cause sinking; compressed air forces water out to decrease density and cause rising. The submarine can achieve neutral buoyancy by matching its density to the surrounding water.

Hot-Air Balloons

Archimedes' Principle applies to gases as well as liquids. A hot-air balloon rises because the air inside the balloon is heated, becoming less dense than the cooler surrounding air. The buoyant force (equal to the weight of the displaced cooler air) is greater than the weight of the balloon and its payload, causing the balloon to rise.

Factor	Cold Air (Outside)	Hot Air (Inside Balloon)
Density	Higher	Lower
Weight of displaced air	Greater	(not applicable)
Buoyant force	Upward	—
Result	Balloon rises	—

Hydrometers

A **hydrometer** is an instrument used to measure the density of liquids. It floats upright in a liquid, and the depth to which it sinks indicates the liquid's density. In denser liquids, the hydrometer floats higher; in less dense liquids, it sinks deeper. Hydrometers are used to test battery acid, measure alcohol content in beverages, and check the salinity of water.

Life Jackets and Floating Devices

Life jackets are designed to increase the volume of a person without significantly increasing their mass. By adding buoyant material (such as foam), the person's average density becomes less than water, ensuring they float even if unconscious.

Fish and Buoyancy

Many fish have a swim bladder — an internal gas-filled organ that can be inflated or deflated to adjust buoyancy. By changing the volume of gas in the swim bladder, fish can rise or sink in the water column without expending energy on swimming.

Part 6: Comparing Pascal's Principle and Archimedes' Principle

Aspect	Pascal's Principle	Archimedes' Principle
Focus	Pressure transmission in confined fluids	Buoyancy and floating
Key concept	Pressure applied to confined fluid is transmitted equally in all directions	Buoyant force equals weight of displaced fluid
Primary application	Hydraulic systems (car jacks, brakes, presses)	Shipbuilding, submarines, hydrometers

Aspect	Pascal's Principle	Archimedes' Principle
Equation	$P = F / A$ (pressure is constant throughout)	$F_b = \rho_{\text{fluid}} \times V_{\text{displaced}} \times g$
Type of fluid	Confined (enclosed) fluid	Any fluid (liquid or gas)
Example	Hydraulic lift lifting a car	Ship floating on water

Part 7: Problem-Solving with Archimedes' Principle

Sample Problem 1: Calculating Buoyant Force

Problem: A rock has a volume of 0.02 m^3 and is completely submerged in water (density = 1000 kg/m^3). What is the buoyant force acting on the rock? (Use $g=9.8 \text{ m/s}^2$)

Solution:

$$F_b = \rho_{\text{water}} \times V_{\text{displaced}} \times g = 1000 \text{ kg/m}^3 \times 0.02 \text{ m}^3 \times 9.8 \text{ m/s}^2 = 196 \text{ N}$$

Sample Problem 2: Determining if an Object Floats

Problem: A wooden block has a mass of 0.5 kg and a volume of 0.0008 m^3 . Will it float in water? (Density of water = 1000 kg/m^3)

Step 1: Calculate the density of the block.

$$\rho_{\text{block}} = m / V = 0.5 \text{ kg} / 0.0008 \text{ m}^3 = 625 \text{ kg/m}^3$$

Step 2: Compare with density of water.

$$625 \text{ kg/m}^3 < 1000 \text{ kg/m}^3$$

Conclusion: The block is less dense than water, so it will float.

Key Terms

Term	Definition
Buoyancy	The upward force exerted by a fluid that opposes the weight of an immersed object
Buoyant force	The upward force that causes objects to feel lighter in fluids
Archimedes' Principle	The buoyant force on an object equals the weight of the fluid displaced by the object
Density	Mass per unit volume; a measure of how tightly packed matter is
Displacement	The volume of fluid pushed aside when an object is submerged
Hydrometer	An instrument that measures the density of liquids using buoyancy
Swim bladder	A gas-filled organ in fish that adjusts buoyancy
Neutral buoyancy	A state where an object's density equals the fluid's density; the object remains suspended at any depth

Review Questions

Responses should be written on a separate sheet of paper.

1. State **Archimedes' Principle** in your own words. What is the relationship between the buoyant force and the weight of displaced fluid?
2. A rock weighs 15 N in air and 10 N when fully submerged in water. Calculate the buoyant force acting on the rock. What does this value represent?
3. Why does a steel ship float while a solid steel nail sinks? Explain using the concepts of density and displacement.
4. Ice has a density of 0.92 g/cm^3 , while seawater has a density of approximately 1.03 g/cm^3 . What percentage of an iceberg would you expect to be submerged? Explain your reasoning.
5. Describe two real-world applications of Archimedes' Principle other than ships and submarines. For each application, explain how the principle is applied.